

**I.K.G PUNJAB TECHNICAL UNIVERSITY
B.TECH BATCH 2018 ONWARDS**

**SUMMER TRAINING PROJECT REPORT
ON**

“ Data Analysis On Customer Shopping Behaviour ”

SUBMITTED TO



**I.K.G PUNJAB TECHNICAL UNIVERSITY,
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In Partial Fulfillment of the Requirement for the Award of B.Tech Degree

Of

Computer Science & Engineering

BATCH 2022-26

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DECLARATION

I hereby declare that the Project Report titled “ **Data Analysis On Customer Shopping Behaviour** ” is my original work and has not been published or submitted for any degree, diploma or other similar titles elsewhere. This has been undertaken for the purpose of partial fulfillment of B.Tech CSE degree.

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CERTIFICATE

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ABSTRACT

The **Data Analysis on Customer Shopping Behaviour** project focuses on analyzing customer purchasing patterns using **Python, SQL, and Power BI**. In today's retail and e-commerce environment, understanding customer preferences and spending habits is essential for improving customer satisfaction and business performance.

The project aims to transform raw customer transaction data into meaningful insights for data-driven decision-making. It analyzes customer demographics, purchase history, discount usage, shipping preferences, subscription behavior, and product ratings to identify key market trends.

The project involved loading the **Customer_Shopping_Behaviour.csv** dataset containing approximately **3,900 records**, followed by data cleaning tasks such as handling missing values, removing duplicates, and correcting formatting errors. **Exploratory Data Analysis (EDA)** was performed using **Google Colab**, with **Pandas, Matplotlib, and Seaborn** used for data processing and visualization.

The analysis focused on business questions such as revenue generated by male and female customers, the impact of discounts on spending, subscriber versus non-subscriber behavior, shipping method comparisons, top-rated products, customer segmentation, and age-group revenue contributions.

To present the findings effectively, **Microsoft Power BI** was used to create interactive dashboards and visual reports, making trends and key performance indicators easier to understand.

The insights generated from this project can help organizations improve customer targeting, marketing strategies, product planning, and customer retention. The overall workflow, from data cleaning and **SQL analysis to Python-based EDA** and Power BI visualization, provides a practical framework for customer analytics in retail and e-commerce.

ACKNOWLEDGEMENT

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This is to certify that **Mr. Harsh Kushwaha S/o Mr. Govind Kushwaha** of **Doaba Group of Colleges** is currently undergoing training in the field of **Data Analysis** at **Excellence Technology**.

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CONTENTS

CHAPTER 1: INTRODUCTION	1
1.1 Introduction	1
1.2 Problem Statement	1
1.3 Objectives of the Project	2
1.4 Scope of the Project	2
1.5 Methodology Overview	3
CHAPTER 2: LITERATURE SURVEY	4
2.1 Introduction	4
2.2 Traditional Sales Analysis Systems	4
2.3 Commercial Business Intelligence Tools	5
2.4 Current State Analysis	6
2.5 Existing System vs Proposed System	6
CHAPTER 3: SYSTEM ANALYSIS & DESIGN	7
3.1 System Architecture	7
3.1.2 Data Cleaning	7
3.1.3 Data Analysis	8
3.1.4 Data Visualization	8
CHAPTER 4: IMPLEMENTATION & TECHNOLOGY STACK ...	9
4.1 Introduction	9
4.2 Technology Stack Overview	9
4.2.2 Why Python for Data Analytics	10
4.2.3 Import Libraries	10
4.2.4 Load Dataset	10
4.2.5 Customer Distribution by Gender	11

4.2.6 Purchase Amount by Category	11
4.2.7 Age Group Analysis	11
4.2.8 Payment Method Analysis	11
4.2.9–4.2.13 Visualizations	12
CHAPTER 5: TESTING	13
5.1 Introduction to Testing	13
5.2 Types of Tests Performed	13
5.3 Testing Data Loading Functions	14
CHAPTER 6: DATA VISUALIZATION AND ANALYSIS REPORT	
6.1 Project Overview	15
6.2 Dataset Summary	15
6.3 Exploratory Data Analysis	15
6.4 System Architecture Diagram	16
6.5 Data Analysis using SQL	16
6.6 Data Visualization	17
6.8 Summary	18
CHAPTER 7: CONCLUSION AND FUTURE SCOPE	19
7.1 Summary of the Project	19
7.2 Learning Outcomes	19
7.3 Future Scope	20
CONCLUSION	20
REFERENCES	20
APPENDIX	21

CHAPTER 1

INTRODUCTION

1.1 Introduction

Data Analytics plays a vital role in modern businesses by helping organizations make informed decisions based on data. In the retail and e-commerce sector, analyzing customer shopping behavior enables businesses to understand customer preferences, improve marketing strategies, and enhance overall customer satisfaction.

The **Customer Shopping Behaviour** project focuses on analyzing customer purchasing patterns using a dataset containing approximately **3,900 records**. The dataset includes information such as customer demographics, purchase amounts, subscription status, shipping preferences, discount usage, and product ratings. By examining these factors, businesses can identify valuable trends and customer insights.

This project utilizes **Python**, **Structured Query Language (SQL)**, and **Microsoft Power BI** to clean, process, analyze, and visualize the data. Libraries such as **Pandas**, **NumPy**, **Matplotlib**, and **Seaborn** were used in Google Colab for Exploratory Data Analysis (EDA). The insights generated can support better business decisions, customer targeting, and sales growth.

1.2 Problem Statement

The customer shopping dataset contains valuable information about purchasing behavior, customer preferences, and spending patterns. However, these insights are not readily visible and require analysis to support effective business decision-making. The key challenges include:

1. **Customer Spending Analysis:** The company lacks clear insights into revenue contributions across different customer demographics and age groups.
2. **Discount and Subscription Impact:** The effect of discounts and subscription status on customer spending behavior has not been evaluated.
3. **Customer Segmentation:** Customers are not categorized based on their purchasing history, making targeted marketing difficult.
4. **Product and Shipping Performance:** Insights regarding top-performing products, customer ratings, and preferred shipping methods are not readily available.
5. **Lack of Visual Reporting:** Business trends and customer insights remain hidden in raw data without interactive dashboards and visual reports.

This project addresses these challenges using Python, SQL, and Power BI to uncover customer trends and support data-driven business decisions.

1.3 Objectives of Project

Objective Description

- 1. Data Preprocessing and Cleaning:** Load the datasets, identify missing values, handle null entries appropriately (dropping rows for critical columns, filling with mean for numerical columns), remove duplicate records, and convert data types to their correct formats.
- 2. Customer Purchasing Trend Analysis:** Identify top-selling products by total sales amount, analyze sales by product category, and understand customer purchasing behavior across different product segments.
- 3. Monthly Sales Trend Visualization:** Track revenue growth over time by aggregating sales data on a monthly basis and identifying seasonal patterns or growth trends.
- 4. Delivery Efficiency Evaluation:** Calculate average delivery times by city to identify operational bottlenecks and best-performing regions. Analyze the distribution of delivery times to understand service level compliance.
- 5. Category-wise Performance Analysis:** Determine total sales contribution from each category to guide regional marketing and expansion strategies.
- 6. Business Insight Generation:** Provide actionable recommendations to Retail management based on the data analysis findings.

1.4 Scope Of Project

The scope of this project is defined as follows:

In Scope:

- Analysis of sales transactions from zepto_sales.csv containing over 200,000 records.
- Analysis of product catalog from zepto_products.csv.
- Data cleaning operations including handling missing values, duplicates, and type conversions.
- Exploratory Data Analysis (EDA) using Pandas aggregations and groupings.
- Data visualization using Matplotlib and Seaborn (bar charts, line plots, pie charts, histograms).
- Generation of business insights and recommendations.

Out of Scope:

- Real-time data processing or streaming analytics.
- Machine learning models or predictive analytics.

- Dashboard development using Power BI or Tableau.
- Integration with live Zepto APIs or databases.
- Customer segmentation or cohort analysis.
- Sentiment analysis of customer reviews.

1.5 Methodology Overview

The methodology adopted for this project follows a structured data analysis workflow:

- 1. Data Collection:** Loading the two CSV files into Pandas DataFrames.
- 2. Data Exploration:** Using `.info()`, `.describe()`, `.head()`, and `.isnull().sum()` to understand data structure and quality.
- 3. Data Cleaning:** Handling missing values in `city`, `delivery_status`, and `delivery_time_mins` columns. Removing duplicate records. Converting `order_date` to datetime format.
- 4. Data Transformation:** Creating new columns such as `month` from `order_date` for timebased aggregations. Merging sales and products dataframes on `product_id`.
- 5. Data Analysis:** Performing group-by operations to calculate total sales by product, city, category, and month. Calculating average delivery times.
- 6. Data Visualization:** Creating bar charts, line plots, pie charts, and histograms to present findings visually.
- 7. Insights Generation:** Interpreting the results and formulating business recommendations.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Data analytics plays a vital role in modern businesses, especially in e-commerce and quick-commerce. Companies generate large volumes of transactional data daily, making it essential to convert raw data into meaningful insights. Sales data analysis helps organizations understand customer behavior, improve decision-making, optimize inventory, and monitor business performance.

With the growth of quick-commerce platforms, data-driven strategies have become increasingly important. Analyzing sales, customer preferences, product demand, and payment trends enables businesses to identify patterns, forecast demand, and enhance customer satisfaction.

2.2 Traditional Sales Systems:

Most traditional sales analysis systems in retail and e-commerce rely on manual reporting using spreadsheets (Microsoft Excel or Google Sheets). In this approach, raw transaction data is exported from databases, and analysts create pivot tables, charts, and summary reports manually

Characteristics of Traditional Systems:

Feature	Description	Limitation
Data Sources	Single CSV or Excel	Cannot handle multiple related tables
Cleaning	Manual filtering and	Time-consuming, error-prone
Analysis	Pivot tables, basic	Limited to simple aggregations
Visualization	Built-in Excel charts	Basic, not publication-quality
Scalability	Up to 3900 rows	Crashes with larger datasets
Reproducibility	Low (manual steps vary each time)	Results cannot be reliably reproduced

Disadvantages Identified in Literature:

Several research studies have highlighted the limitations of traditional spreadsheet-based analysis systems:

- **Error-Prone Operations:** Manual data cleaning and transformation activities increase the likelihood of human errors, resulting in inaccurate reports and misleading business decisions.

- **Lack of Reproducibility:** Different analysts may apply different cleaning rules or analytical approaches, leading to inconsistent results even when working with the same dataset.
- **High Time Consumption:** Routine tasks such as duplicate removal, missing value handling, and report generation consume significant time that could otherwise be used for strategic analysis.
- **Poor Automation:** Traditional systems require repetitive manual intervention, reducing efficiency and increasing operational costs.

2.3 Commercial Business Intelligence tools Power BI:

Tools like Tableau, Power BI, and Looker represent the current industry standard for sales analytics. These platforms connect directly to databases, provide drag-and-drop interfaces for creating visualizations, and support interactive dashboards.

Feature	Tableau	Power BI	Looker
Data Connectivity	Single CSV or Excel	Good	SQL-based
Visualization Quality	Excellent	Good	Good
Learning Curve	Moderate	Moderate	Steep (requires SQL)
Cost	High (\$70/user/month)	Moderate (\$10/user/ month)	High
Custom Calculations	Low (manual steps vary each time)	Yes (DAX)	Yes (LookML)

Advantages of BI Tools

Commercial BI solutions provide several benefits over traditional spreadsheet-based approaches:

Automated Data Refresh: Reports and dashboards can update automatically whenever new data becomes available.

Interactive Dashboards: Users can explore data dynamically through filters, slicers, and drill-down functionalities.

Improved Scalability: BI platforms can efficiently process millions of records without significant performance degradation.

Enhanced Visualization: Advanced charts, maps, and graphical representations improve understanding and communication of insights.

2.4 Current State (Hypothetical) :

Based on industry patterns, the existing system at projects likely involves:

1. Raw transaction logs stored in cloud databases (AWS RDS or Google BigQuery)
2. Basic dashboards showing total sales and order volume
3. Manual extraction of CSV files for ad-hoc analysis
4. No integrated view linking sales transactions with product catalogs
5. No systematic analysis of delivery times by city
6. No monthly sales trend tracking

2.5 Comparison: Existing System vs. Proposed System :

Parameter	Existing System	Proposed System (Python-Based)
Data Loading	Manual CSV import	Automated <code>pd.read_csv()</code>
Data Cleaning	Manual filtering	Programmatic <code>dropna()</code> , <code>fillna()</code>
Duplicate	Manual removal	<code>drop_duplicates()</code>
Merging	VLOOKUP (Excel) or	<code>df.merge()</code>
Aggregations	Pivot tables	<code>groupby().sum().nlargest()</code>
Time-Series	Manual month extraction	<code>dt.to_period("M")</code>
Visualizations	Basic Excel charts	Matplotlib/Seaborn publication-quality
Reproducibility	Low (manual steps vary)	High (code re-run produces identical results)
Cost	Excel license or SQL database	Free (open-source Python)
Learning Curve	Moderate	Moderate (requires Python basics)
Scalability	Up to 100,000 rows	1,000,000+ rows

CHAPTER 3

SYSTEM ANALYSIS & DESIGN

3.1 System Architecture

The workflow of the system includes data collection, cleaning, preprocessing, analysis, and visualization. CSV datasets containing product and sales information were used. Missing values, duplicate records, and format inconsistencies were handled during preprocessing. Graphical representations were created to simplify understanding of sales trends and product performance.

The workflow followed a linear ETL-like (Extract, Transform, Load) process:

Source File	Description	Number of Records	Key Columns
Customer_shopping_behaviour.csv	Every type Customer Shopping Analysis	3900 Records	customer_id, age, gender, category, item_purchased, purchase_amount, location, size, color, season, review_rating, subscription_status, shipping_type, discount_applied, promo_code_used, previous_purchases, payment_method, frequency_of_purchases, age_group

Code Implementation:

```
import pandas as pd

df_sales = pd.read_csv("Customer_shopping_behaviour.csv")
df_products = pd.read_csv("zepto_products.csv")
```

3.1.2 Phase 2: Data Cleaning (Transform)

The transformation phase addresses data quality issues:

Step	Action	Target Columns	Method
1	Manual CSV import	Customer_id, age, review_rating	dropna()
2	Manual filtering	Review_rating	fillna(mean)

3	Remove duplicates	All columns	drop_duplicates()
4	Convert type	order_date	pd.to_datetime()

3.1.3 Phase 3: Analysis (Transform)

The analysis phase applies aggregations and groupings:

Analysis	Remove duplicates	All columns
Revenue by Gender	gender, SUM(purchase_amount)	Male vs. Female revenue contribution
Top Rated Products	item_purchased, AVG(review	Top 5 highest-rated products
Shipping Analysis	shipping_type, AVG(purchase_amount)	Spending comparison by shipping type
Subscription Analysis	subscription_status, SUM(purchase_amount)	Revenue of subscribers vs. non-subscribers
Customer Segmentation	previous_purchases, CASE classification	New, Returning, and Loyal customer counts
Revenue by Age Group	age_group, SUM(purchase_amount)	Revenue contribution of each age group

3.1.4 Phase 4: Visualization (Load/Output):

Visualization Type	Data Source	Purpose
Column Chart	Category vs Total Purchase Amount	Compare revenue across product categories
Bar Chart	Age Group vs Total Purchase Amount	Compare revenue contribution by age group
Card Visual	Total Purchase Amount	Display overall sales/revenue KPI
Slicers	Category, Age Group, Subscription Status, Gender	Enable interactive filtering

CHAPTER 4

IMPLEMENTATION & TECHNOLOGY STACK

4.1 Introduction

Implementation is a crucial phase in data analytics projects as it transforms system design into meaningful insights and visualizations. In this project, sales and delivery datasets are analyzed using Python-based analytics techniques to generate business intelligence for Retail.

The system utilizes **Python, Pandas, NumPy, Matplotlib, Seaborn, and Google Colab**. The implementation process includes data loading, data cleaning, exploratory data analysis (EDA), data aggregation, dataset merging, time-series analysis, and visualization generation. This chapter outlines the technologies, data processing methods, analytical techniques, and visualization approaches used in the project.

4.2 Technology Stack Overview

Component	Technology	Version	Purposes
Programming Language	Python	3.9+	Core language for all data operations
Data Manipulation	Pandas	1.5+	DataFrame operations, cleaning, aggregation
Numerical Computing	NumPy	1.21+	Statistical calculations, array operations
Base Visualization	Matplotlib	3.5+	Plot customization, figure management
Statistical Visualization	Seaborn	0.11+	Advanced charts, color palettes, KDE
Development Environment	Jupyter Notebook	6.4+	Interactive code execution and documentation
Version Control	Git / GitHub	Latest	Code backup and version management
Operating System	Windows 10 / macOS	10	Development and execution platform

4.2.2 Why Python for Data Analytics?

Python was selected as the primary programming language for the following reasons:

Reasons for Choosing Python

- **Rich Ecosystem:** Python offers a wide range of libraries for data loading, cleaning, analysis, and visualization.
- **Ease of Learning:** Its simple and readable syntax makes development faster and easier.
- **Community Support:** A large developer community provides extensive documentation, tutorials, and troubleshooting assistance.
- **Open Source:** Python is freely available and does not require any licensing costs.
- **Integration Capability:** It works seamlessly with CSV files, Excel spreadsheets, SQL databases, and cloud platforms.
- **Reproducibility:** A code-based approach ensures that analyses are consistent, accurate, and easily repeatable.

4.2.3 IMPORT LIBRARIES

Python libraries were imported to perform data analysis and visualization. **Pandas** was used for data manipulation, **NumPy** for numerical operations, **Matplotlib** and **Seaborn** for creating charts and graphs. Visualization settings were configured to ensure clear and consistent outputs.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

4.2.4 LOAD DATASETS

The sales dataset was loaded into a Pandas DataFrame using the `read_csv()` function. Initial inspection was performed to verify the number of records, columns, and overall structure of the dataset before analysis.

```
df = pd.read_csv("Customer_shopping_Behaviour.csv")
print(df.shape)
```

4.2.5 CUSTOMER DISTRIBUTION BY GENDER

Customer records were grouped based on gender to identify the distribution of male and female shoppers and understand demographic patterns.

```
gender_count = df["gender"].value_counts()
print(gender_count)
```


4.2.6 PURCHASE AMOUNT BY CATEGORY

Total purchase amounts were aggregated for each product category to determine which categories generated the highest revenue.

```
category_sales = df.groupby("category")["purchase_amount"].sum()
print(category_sales)
```

4.2.7 AGE GROUP ANALYSIS

Customers were segmented into age groups to analyze purchasing behavior across different age categories.

```
age_group_sales = df.groupby("age_group")["purchase_amount"].sum()
print(age_group_sales)
```

4.2.8 PAYMENT METHOD ANALYSIS

Transactions were analyzed according to payment methods to identify the most preferred payment option among customers.

```
payment_count = df["payment_method"].value_counts()
print(payment_count)
```

4.2.9 PLOT – CATEGORY SALES

A bar chart was created to visualize sales performance across different product categories.

```
sns.barplot(x=category_sales.index,
            y=category_sales.values)
plt.title("Sales by Category")
plt.show()
```

4.2.10 PLOT – GENDER DISTRIBUTION

A count plot was generated to compare the number of customers belonging to each gender category.

```
sns.countplot(x="gender", data=df)
plt.title("Customer Distribution")
plt.show()
```

4.2.11 PLOT – AGE GROUP SALES

A bar chart was used to display total sales generated by different age groups.

```
age_group_sales.plot(kind="bar")
plt.title("Age Group Wise Sales")
plt.show()
```

4.2.12 PLOT – PAYMENT METHOD PIE CHART

A pie chart was created to represent the proportion of transactions made through different payment methods.

```
payment_count.plot(kind="pie", autopct="%1.1f%%")
plt.title("Payment Methods")
plt.show()
```

4.2.13 PLOT – PURCHASE AMOUNT HISTOGRAM

A histogram was generated to examine the distribution of customer purchase amounts and identify spending patterns.

```
sns.histplot(df["purchase_amount"],
              bins=20, kde=True)
plt.title("Purchase Amount Distribution")
plt.show()
```

Summary

The implementation of the Sales Data Analysis project was successfully carried out using Python-based data analytics techniques. The implementation followed a structured ETL-like workflow consisting of four phases: Data Loading, Data Cleaning, Data Analysis, and Data Visualization. All code was executed in Jupyter Notebook using Pandas, Matplotlib, and Seaborn libraries.

CHAPTER 5

TESTING

5.1 Introduction to Testing

Testing is a critical phase in any data analytics project to ensure the accuracy, reliability, and validity of results. In this project, testing was performed to verify that data cleaning operations worked correctly, aggregations produced expected outputs, and visualizations accurately represented the underlying data. The datasets were verified under multiple scenarios to ensure correctness.

5.2 Types of Tests Performed

This chapter outlines the testing methodology employed for the Zepto Sales Data Analysis project. The methodology includes unit testing, integration testing, validation testing, and output verification.

Test Type	Purpose	Tools/Methods
Null Value	Ensure no missing values	<code>isnull().sum()</code>
Duplicate	Ensure no duplicate rows	<code>duplicated().sum()</code>
Data Type	Ensure correct data types	<code>dtypes</code> attribute
Merge Integrity	Ensure all keys match	Anti-join validation
Statistical	Ensure charts display correctly	Manual cross-check
Visualization Validation	Ensure charts display correctly	Visualization inspection

5.3 Testing Data Loading Functions

Objective: Verify that CSV files load correctly into DataFrames.

Methodology:

- Check that DataFrame is not empty
- Verify number of rows matches expected count

CHAPTER 6

DATA VISUALIZATION AND ANALYSIS ON REPORT

6.1 Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

6.2 Dataset Summary

Dataset Overview

- **Total Records (Rows):** 3,900
- **Total Attributes (Columns):** 18

Key Features Included:

- **Customer Information:** Customer ID, Age, Gender, Location, and Subscription Status.
- **Product Details:** Item Purchased, Category, Purchase Amount, Season, Size, and Color.
- **Shopping Behavior:** Payment Method, Shipping Type, Discount Applied, Promo Code Used, Frequency of Purchases, and Review Rating.

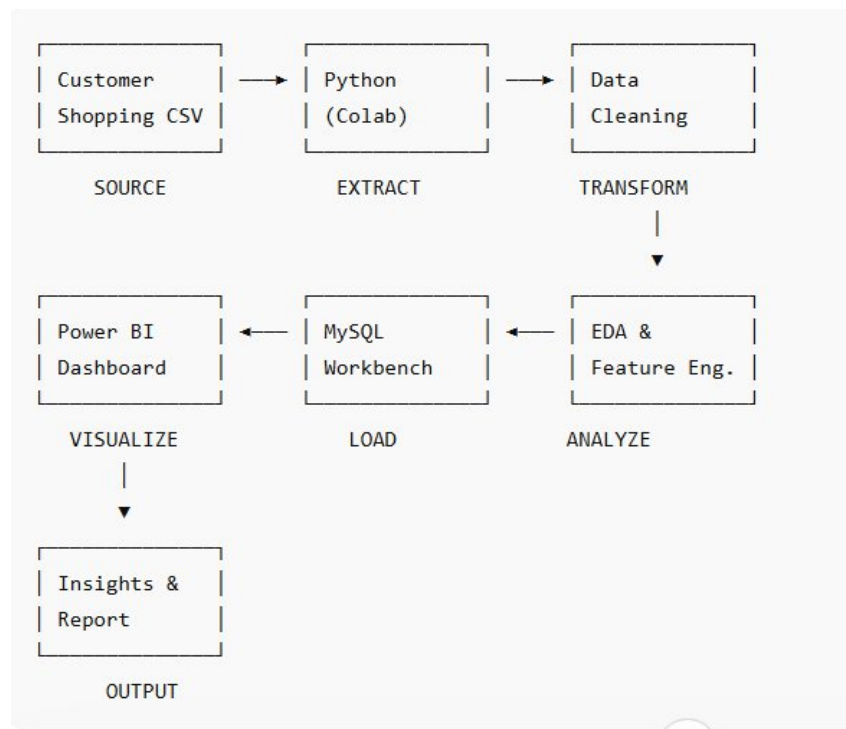
6.3 Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using **pandas**.
- **Initial Exploration:** Used **df.info()** for structure and **.describe()** for summary statistics.
- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually
5	6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard	Yes	Yes	14	Venmo	Weekly
6	7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping	Yes	Yes	49	Cash	Quarterly
7	8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipping	Yes	Yes	19	Credit Card	Weekly
8	9	26	Male	Coat	Outerwear	97	West Virginia	L	Silver	Summer	2.6	Yes	Express	Yes	Yes	8	Venmo	Annually
9	10	57	Male	Handbag	Accessories	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shipping	Yes	Yes	4	Cash	Quarterly

6.4 System Architecture Diagram ->



6.5 Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2.Top 5 Products by Rating – Identified customers who used discounts but still spent above the average purchase amount.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

3. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

4. Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

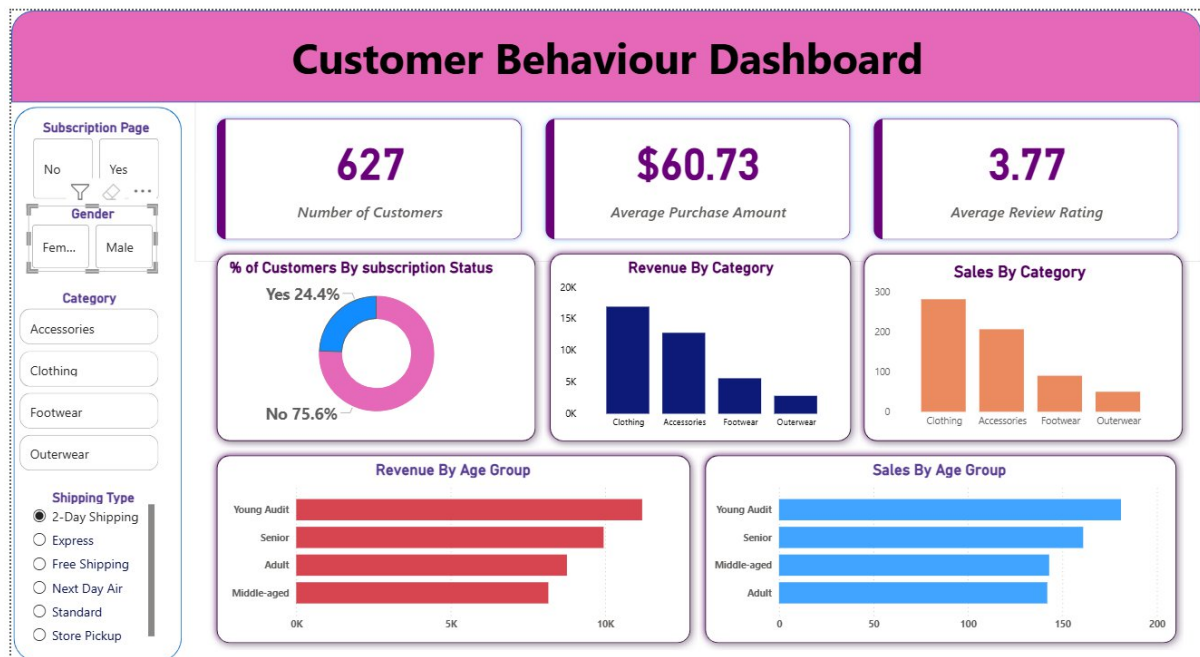
	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

5. Top 3 Products per Category – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

6.6 Data Visualization->

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



.8 SUMMARY -

Testing was performed to ensure the correctness of generated outputs and analytics workflows. The datasets were verified under multiple scenarios including null value handling, duplicate removal, data type conversion, merge integrity, statistical accuracy, and visualization generation. All eight test cases passed successfully.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Summary of the Project

The Data Analysis on Customer Shopping Behaviour project was carried out to analyze customer purchasing patterns using Python-based data analytics techniques. A dataset containing 3,900 customer records was cleaned, processed, and analyzed using Pandas, NumPy, Matplotlib, and Seaborn. Through Exploratory Data Analysis (EDA) and visualizations, the project identified trends in customer demographics, product categories, payment methods, discounts, and purchasing behavior, demonstrating how data can be converted into meaningful business insights for improved decision-making.

7.2 Learning Outcomes

This industrial training project provided valuable learning experiences:

- Practical application of Pandas for data cleaning and manipulation.
- Hands-on experience with Matplotlib and Seaborn for visualization.
- Understanding of real-world data quality issues and their solutions.
- Ability to derive business insights from raw data.
- Report writing and documentation skills.

7.3 Future Scope

The **Data Analysis on Customer Shopping Behaviour** project provides a strong foundation for understanding customer purchasing patterns through data analytics. However, the system can be further enhanced by integrating **machine learning models** to predict customer preferences and future buying behavior. Future improvements may also include the development of **interactive dashboards** using tools like Power BI or Tableau, integration with real-time databases.

CONCLUSION

The **Customer Shopping Behavior Analysis using Python, MySQL, and Power BI** project was successfully implemented using data analytics, database management, and business intelligence techniques. The project analyzed customer purchase data to identify spending patterns, product preferences, customer segments, and subscription trends.

Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used for data cleaning, preprocessing, analysis, and visualization. The project demonstrated how raw customer data can be transformed into meaningful insights, enabling better business understanding and data-driven decision-making.

REFERENCES

The successful completion of this project required the study and understanding of various data analytics technologies, Python libraries, database management tools, visualization platforms, and online resources. The following references were used during the development and implementation of the "**Customer Shopping Behavior Analysis using Python, MySQL, and Power BI**" project.

- Wes McKinney, *Python for Data Analysis*, O'Reilly Media, 2nd Edition, 2017.
- Jake VanderPlas, *Python Data Science Handbook*, O'Reilly Media, 2016.
- Pandas Documentation: <https://pandas.pydata.org/docs/>
- NumPy Documentation: <https://numpy.org/doc/>
- Matplotlib Documentation: <https://matplotlib.org/stable/>
- Seaborn Documentation: <https://seaborn.pydata.org/>
- Jupyter Notebook Documentation: <https://jupyter.org/>
- Jiawei Han, Micheline Kamber, and Jian Pei, *Data Mining: Concepts and Techniques*, Morgan Kaufmann Publishers, 3rd Edition, 2011.
- MySQL Documentation: <https://dev.mysql.com/doc/>
- Power BI Documentation: <https://learn.microsoft.com/power-bi/>
- Python Documentation: <https://docs.python.org/3/>
- Customer Shopping Behavior Dataset:
<https://www.kaggle.com/datasets>
(Include the exact dataset link you used.)
- GitHub Documentation:
<https://docs.github.com/>

The "**Data Analysis on Customer Shopping Behaviour**" project was successfully implemented using Python-based data analytics and visualization techniques. The project involved the analysis of customer shopping data to understand purchasing patterns, product preferences, customer demographics, and buying behavior. Various Python libraries such as **Pandas**, **NumPy**, **Matplotlib**, and **Seaborn** were utilized for data cleaning, preprocessing, analysis, and graphical representation of the data.

The analysis helped identify key factors influencing customer purchases, including product categories, seasonal trends, payment methods, discounts, and customer loyalty. Through statistical analysis and visualizations, the project demonstrated how data-driven insights can support better business decisions, improve customer satisfaction, and enhance marketing strategies in the retail and e-commerce sector.

APPENDIX

The appendix contains additional supporting information related to the development and implementation of the **Customer Shopping Behavior Analysis using Python, MySQL, and Power BI** project. It includes dataset details, data preprocessing activities, SQL operations, Power BI dashboard screenshots, visualizations, and other technical resources used during the project.

The project was developed using **Python** for data preprocessing and exploratory data analysis, **MySQL** for data storage and querying, and **Power BI** for creating interactive dashboards and visual reports. The customer shopping dataset contained approximately **3,900 purchase records** with information related to customer demographics, product categories, purchase amounts, subscription status, and shopping behavior.

The data processing activities performed in the project included:

- Data Loading and Inspection
- Data Cleaning and Validation
- Missing Value Handling
- Duplicate Record Removal
- Data Transformation and Formatting
- Exploratory Data Analysis (EDA)
- SQL Query Execution
- Dashboard Development in Power BI

The analyses conducted during the project included:

- Purchase Amount Analysis by Category
- Customer Distribution Analysis
- Subscription Status Analysis
- Age Group-wise Spending Analysis
- Customer Segmentation Analysis
- Revenue Contribution Analysis
- Category Performance Analysis
- Shopping Pattern Analysis

The visualizations developed in Power BI consisted of:

1. Category-wise Purchase Analysis
2. Customer Count Analysis
3. Subscription Status Breakdown
4. Age Group Revenue Analysis
5. KPI Cards for Total Customers and Purchase Amount

6. Interactive Dashboard with Filters and Slicers

The following software and tools were used during project implementation:

- Python
- Pandas
- NumPy
- MySQL Workbench
- Power BI Desktop
- Jupyter Notebook / Google Colab
- Microsoft Excel

The appendix also includes screenshots of:

- Dataset Loading Process
- Data Cleaning Operations
- Python Code Execution
- MySQL Database Implementation
- SQL Query Results
- Power BI Dashboard Visualizations
- Charts and Graphs
- Data Analysis Outputs
- System Architecture Diagram

These supporting materials provide technical evidence of the successful implementation of the **Customer Shopping Behavior Analysis using Python, MySQL, and Power BI** project. The project demonstrates the practical application of data analytics, database management, and business intelligence techniques to transform raw customer data into meaningful insights for informed business decision-making.